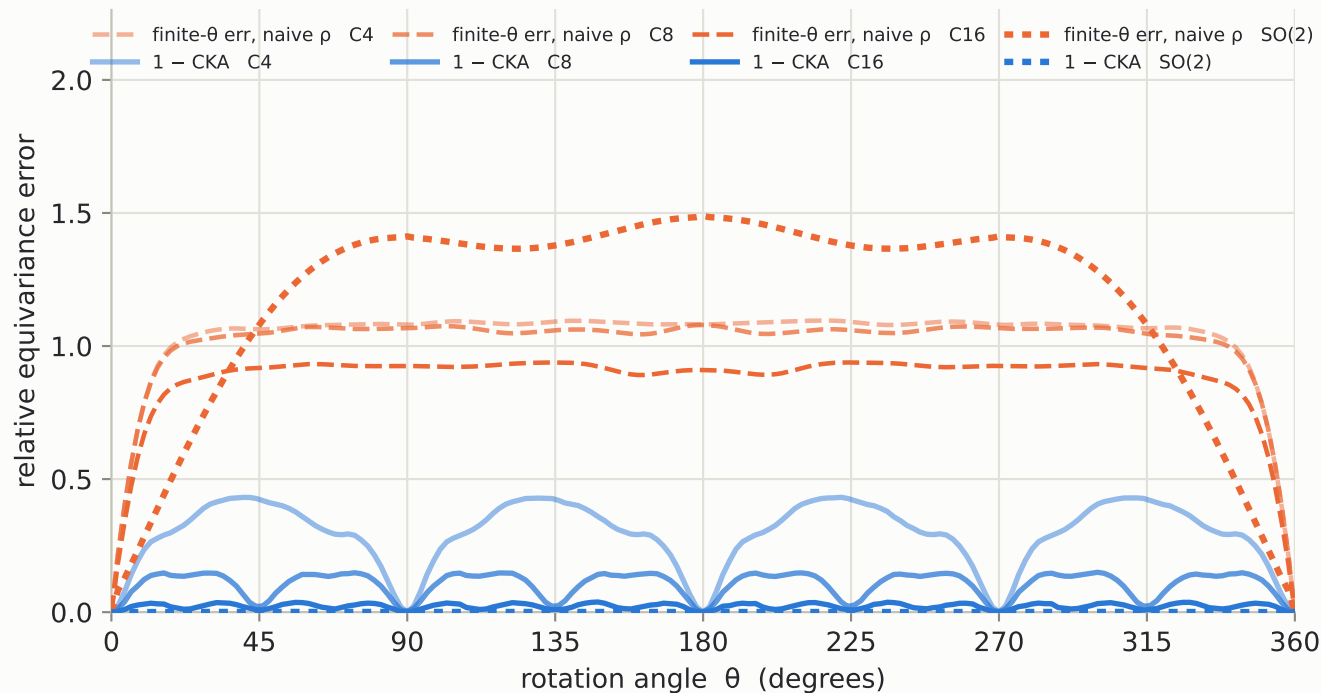


The Lie-derivative family is flat across every group; CKA resolves each one



Only CKA falls with N ($0.522 \rightarrow 0.022$ across $C1 \rightarrow C16$, a factor of 24). The finite- θ error under the naive ρ is flat (0.88–1.06) and inverts at the end: it scores SO(2), the one model equivariant at every θ , worst of all at 1.18. The correct ρ for a regular field also permutes the fiber, which spatial rotation alone misses, so the error saturates near 1 and group order never shows.

LEE is taken at the identity: one scalar per model, no θ argument at all. It separates SO(2) from the rest — which is all it can see — and is flat across the entire C_N family.

LEE (Local Equivariance Error) is that derivative's norm, so it carries units of rad^{-1} while CKA and the finite- θ error are dimensionless ratios: compare its shape across N , not its level against the other two.

what each metric says about group order

